

BLUESTOCK MUTUAL FUND ANALYTICS PLATFORM

Capstone Project Report

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Executive Summary

The Mutual Fund Analytics Platform was developed to provide a comprehensive solution for analysing mutual fund performance, investor behaviour, and market trends. The project covers the complete data lifecycle, including data ingestion, cleaning, storage, analysis, visualization, forecasting, and automation.

The platform integrates multiple mutual fund datasets and transforms them into actionable insights through exploratory data analysis, performance analytics, advanced risk modelling, dashboard development, and recommendation systems. Power BI and Streamlit dashboards were developed to enable interactive exploration of the data, while advanced analytical techniques such as Monte Carlo simulation and portfolio optimization were implemented to support investment decision-making.

The project successfully demonstrates the application of data engineering, business intelligence, and financial analytics techniques in building an end-to-end mutual fund analytics solution.

1. Introduction

Mutual funds have become one of the most popular investment vehicles due to their diversification benefits and professional fund management. However, evaluating mutual fund performance requires the analysis of large volumes of financial and investor data.

This project aims to simplify mutual fund analysis by creating an integrated analytics platform capable of processing historical fund data, calculating

performance metrics, generating insights, and presenting information through interactive dashboards.

2. Problem Statement

Investors often face difficulties in comparing mutual funds, understanding risk levels, evaluating historical performance, and identifying suitable investment opportunities. Traditional analysis methods are time-consuming and require the integration of multiple datasets.

The project addresses these challenges by developing a centralized analytics platform that provides automated data processing, advanced analytics, and interactive visualization capabilities.

3. Project Objectives

The primary objectives of the project are:

- Build an ETL pipeline for mutual fund datasets.
 - Create a structured SQLite database for data storage.
 - Perform data cleaning and validation.
 - Conduct exploratory data analysis.
 - Calculate key mutual fund performance metrics.
 - Develop Power BI and Streamlit dashboards.
 - Implement advanced risk analytics.
 - Build a recommendation engine.
 - Forecast future NAV performance.
 - Automate reporting and data updates.
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4. Data Sources

The project utilizes multiple datasets related to mutual fund operations and investor behaviour.

Datasets Used

- Fund Master Data
- NAV History
- AUM Data by Fund House
- Investor Transactions
- Monthly SIP Inflow Data
- Category Inflows
- Industry Folio Count
- Scheme Performance Count
- Portfolio Holdings
- Benchmark Indices

These datasets were processed and integrated into a unified analytics environment.

5. ETL Pipeline Design

The ETL process was designed to ensure data quality and consistency before analysis.

Workflow

Raw Data → Data Cleaning → Validation → Transformation → SQLite Database

Key ETL Activities

- Data ingestion from source files.
- Handling missing values.
- Removing duplicate records.
- Standardizing formats and categories.
- Data validation checks.
- Loading cleaned datasets into SQLite.

The ETL pipeline enables repeatable and reliable data processing for future updates.

6. Database Design

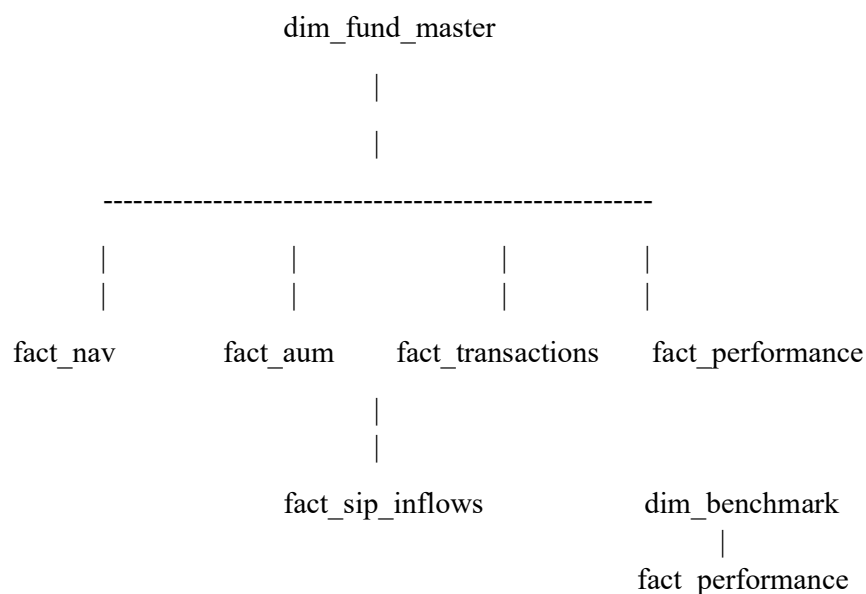
A SQLite database was created to store cleaned datasets and support analytical queries.

Core Tables

- 01_fund_master
- 02_nav_history
- 03_aum_by_fund_house
- 04_monthly_sip_inflows
- 05_category_inflows
- 06_industry_folio_count
- 07_scheme_performance
- 08_investor_transactions
- 09_portfolio_holdings
- 10_benchmark_indices

Design Approach

The database structure enables efficient joins and analysis using common identifiers such as AMFI code and date fields.



7. Data Cleaning and Validation

Several preprocessing techniques were applied to improve data quality.

Data Cleaning Activities

- Missing value treatment.
- Duplicate removal.
- Data type conversion.
- Date formatting.
- Category standardization.
- AMFI code validation.

Validation Checks

- Record count verification.
- Data consistency checks.
- Referential integrity verification.

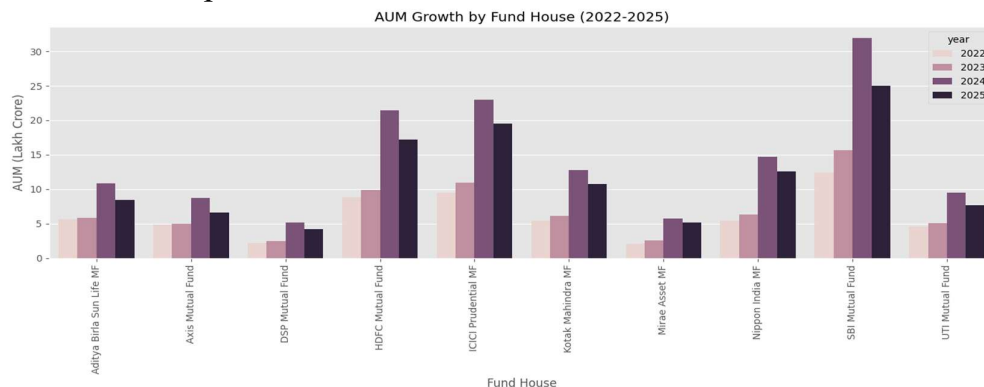
The cleaning process ensured the reliability of subsequent analytical results.

8. Exploratory Data Analysis

EDA was conducted to understand trends, distributions, and relationships within the data.

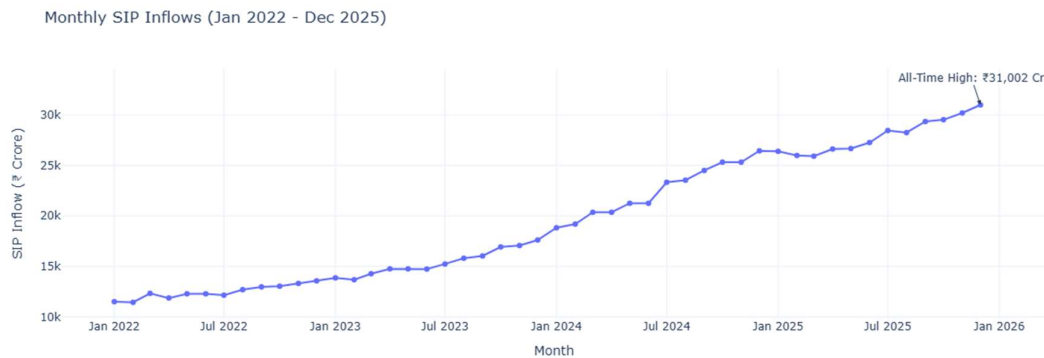
AUM Growth Analysis

The mutual fund industry experienced significant growth in Assets Under Management over the analysis period, indicating increasing investor participation and market expansion.



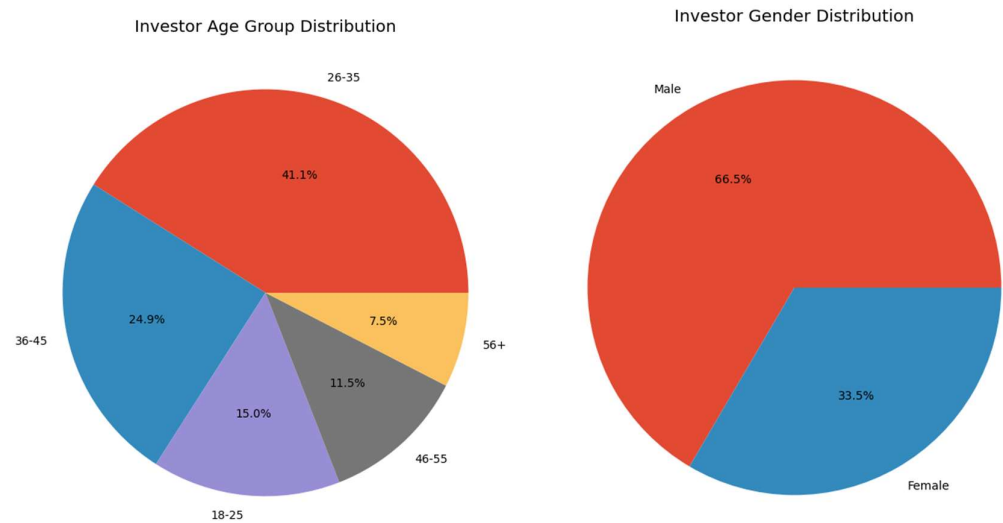
SIP Inflow Analysis

Monthly SIP inflows showed a strong upward trend, reflecting growing investor confidence in systematic investing.



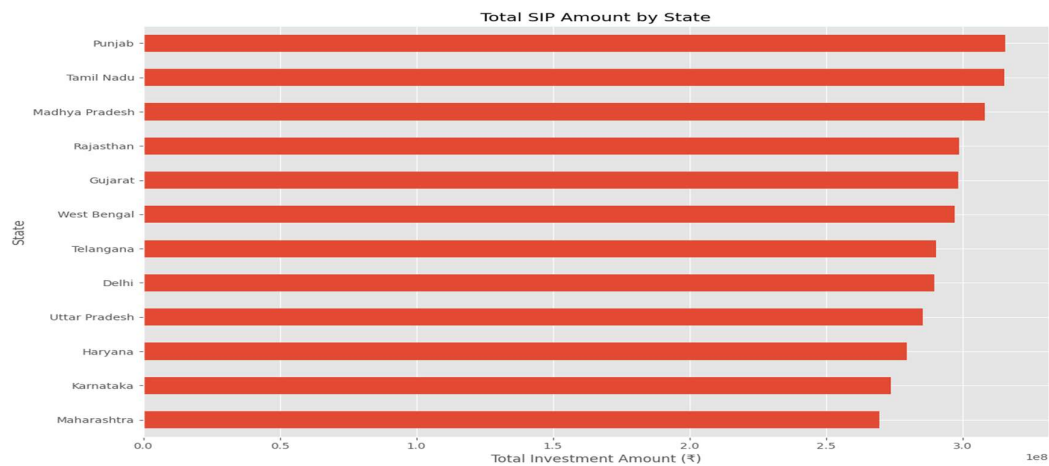
Investor Demographics

Analysis of investor demographics revealed varying participation patterns across age groups and genders.



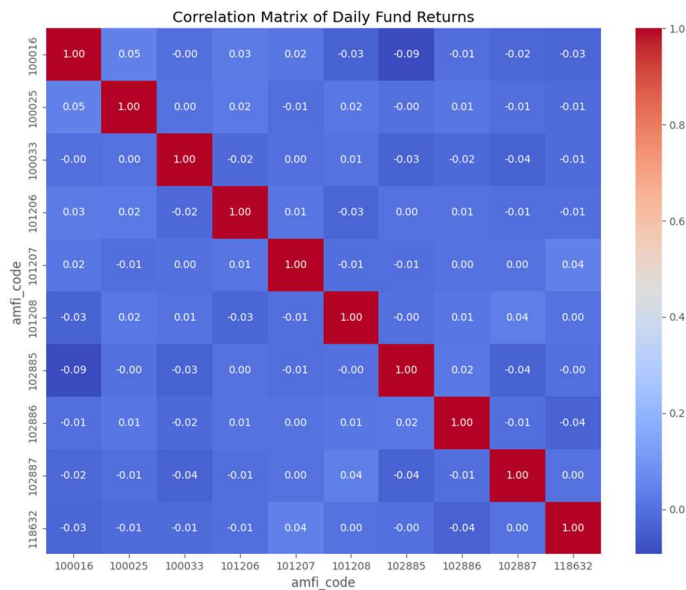
Geographic Distribution

State-wise analysis highlighted regions contributing the highest investment volumes.



Category-wise Fund Flow Analysis

Fund inflows varied significantly across categories, with equity-oriented categories attracting substantial investments.



9. Performance Analytics

Performance analytics were conducted to evaluate fund returns, risk, and benchmark performance.

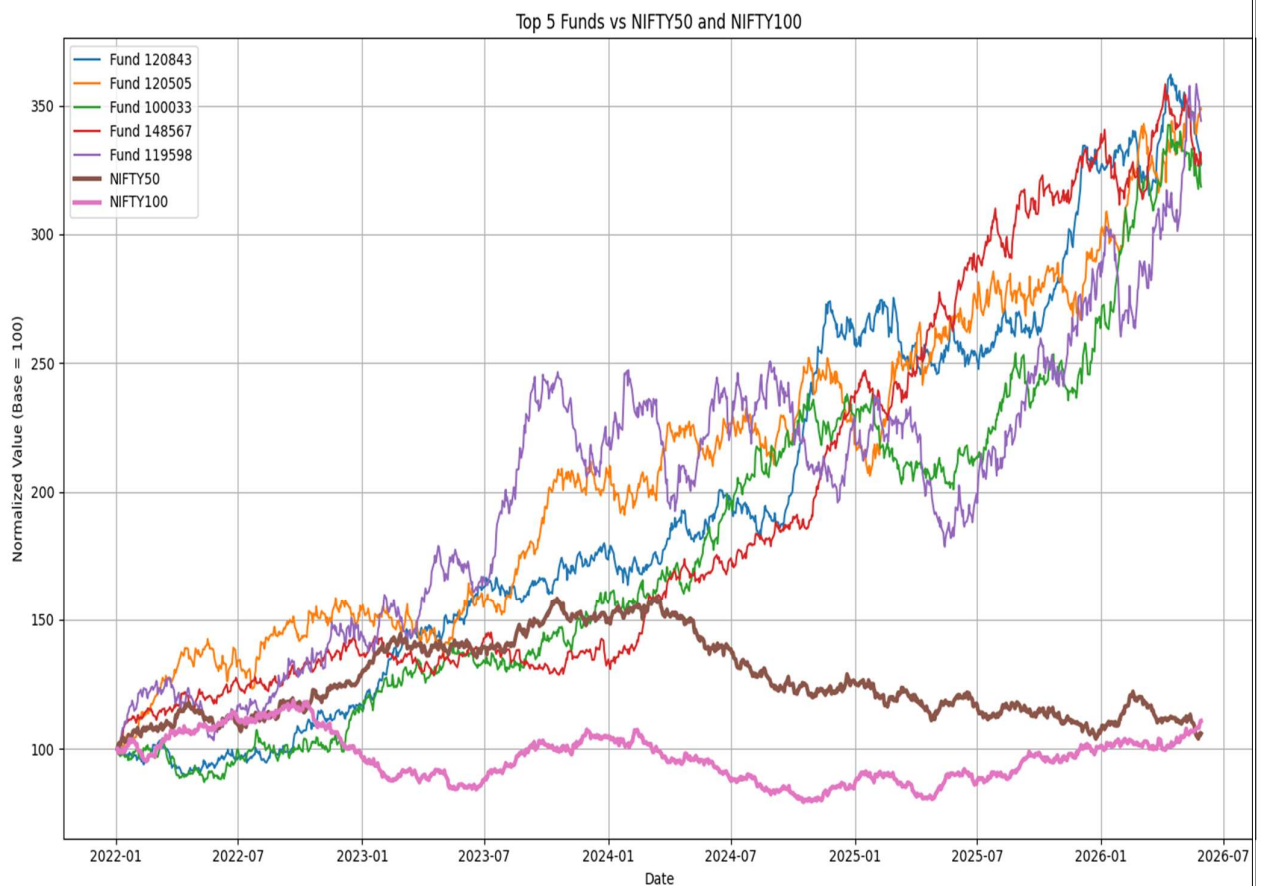
Metrics Calculated

- Daily Returns
- CAGR
- Sharpe Ratio

- Sortino Ratio
- Alpha
- Beta
- Maximum Drawdown

Key Findings

- Several funds outperformed benchmark indices on a risk-adjusted basis.
- Sharpe Ratio rankings identified efficient performers.
- Maximum Drawdown highlighted differences in downside protection.



amfi code	return 3yr pct	expense ratio pct	return rank	expense rank	sharpe rank	alpha	drawdown rank
120843	15.65	1.45	11	19	2	0.273305	13
120505	18.08	1.36	8	14	5	0.292636	25
100033	16.58	1.38	10	15	7	0.271954	20
148567	14.81	1.46	17	20	1	0.269838	8
119598	23.39	1.43	1	18	14	0.30337	36
149323	17.16	1.61	9	30	6	0.265986	22
120504	14.41	0.8	20	11	10	0.211948	12
149324	20.08	1.52	6	22	13	0.300579	37
119094	15.18	1.38	15	15	11	0.260767	28
119599	23.14	0.72	2	5	32	0.048824	40

sharpe score	alpha score	expense score	drawdown score	fund score	return score	overall rank
39	40.7267	22	28	100	30	1
36	40.70736	27	16	99.08	33	2
34	40.72805	26	21	96.82	31	3
40	40.73016	21	33	96.37	24	4
27	40.69663	23	5	93.47	40	5
35	40.73401	11	19	91.06	32	6
31	40.78805	30	29	89.73	21	7
28	40.69942	19	4	87.56	35	8
30	40.73923	26	13	86.82	26	9
9	40.95118	36	1	83.77	39	10

10. Advanced Analytics

Advanced analytical techniques were implemented to provide deeper investment insights.

Value at Risk (VaR) and CVaR

Historical VaR and CVaR were calculated to estimate potential downside risk.

Rolling Sharpe Ratio

Rolling Sharpe analysis was used to evaluate changing risk-adjusted performance over time.

Investor Cohort Analysis

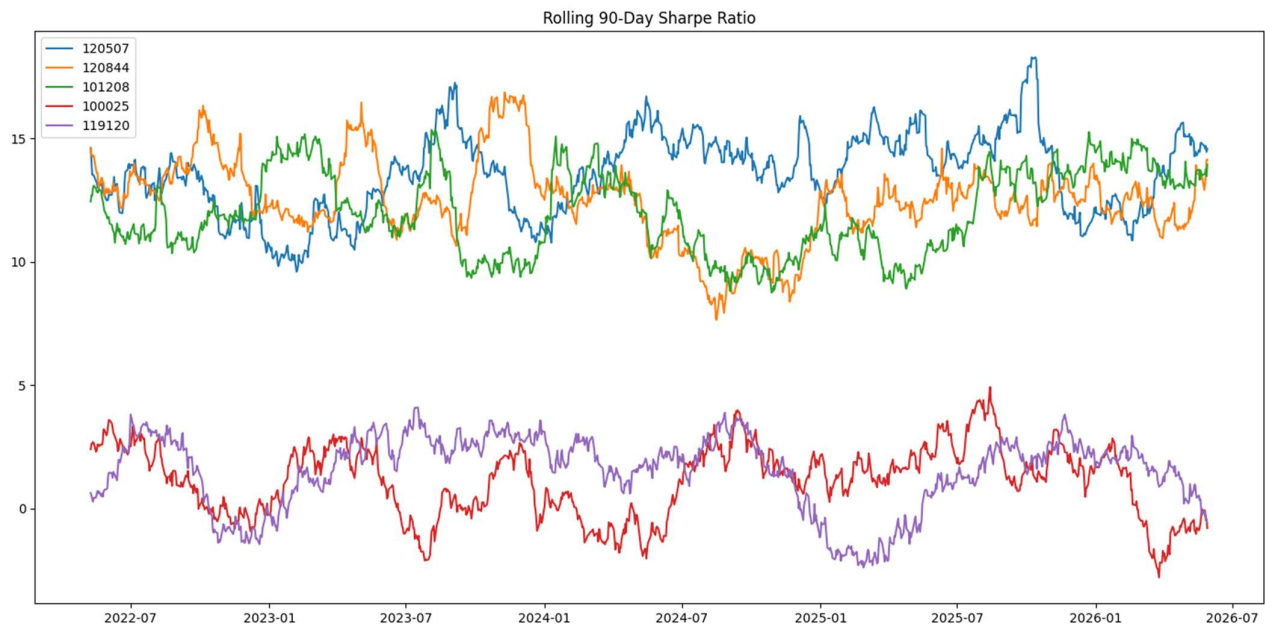
Investor cohorts were analyzed to understand participation trends and investment behavior.

SIP Continuity Analysis

SIP transaction patterns were examined to identify investors at risk of discontinuing investments.

Recommendation Engine

A rule-based recommendation engine was developed using risk profiles and Sharpe Ratio rankings.



11. Monte Carlo Simulation

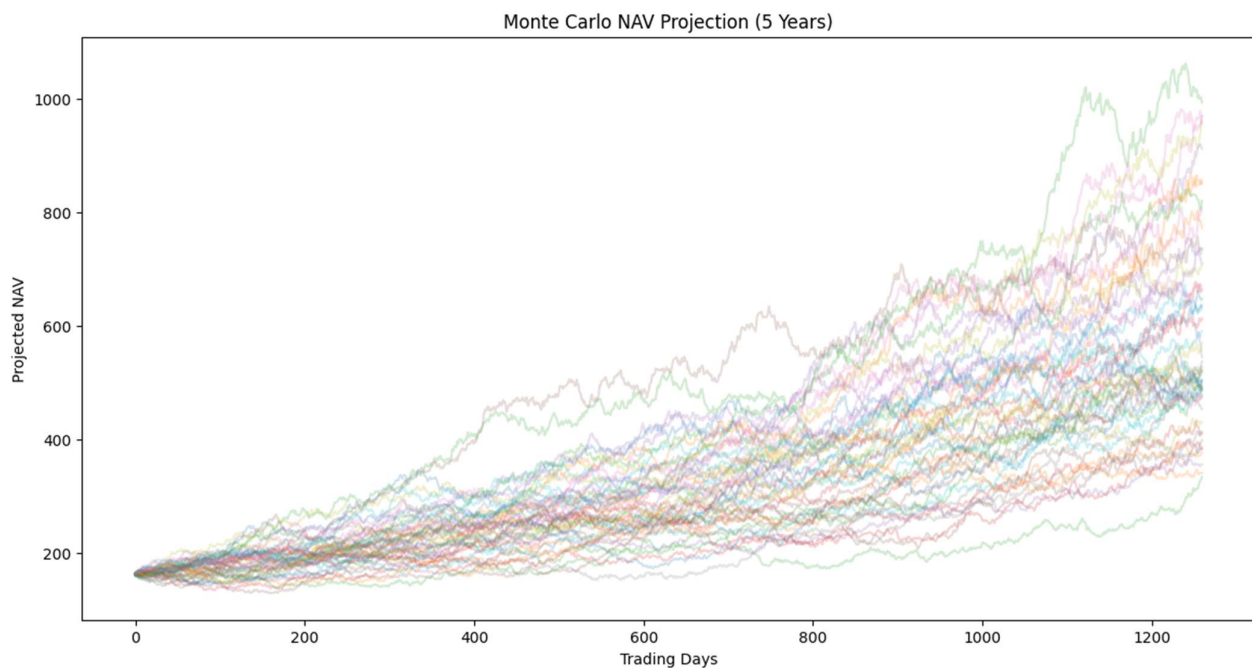
Monte Carlo simulation was implemented to forecast future NAV performance under uncertainty.

Methodology

- Historical returns were used as input.
- 1,000 simulation paths were generated.
- Five-year NAV projections were produced.

Insights

- Expected, optimistic, and pessimistic outcomes were estimated.
- The simulation demonstrated the impact of market volatility on future returns.



12. Portfolio Optimization

Portfolio optimization was performed using the Markowitz Efficient Frontier framework.

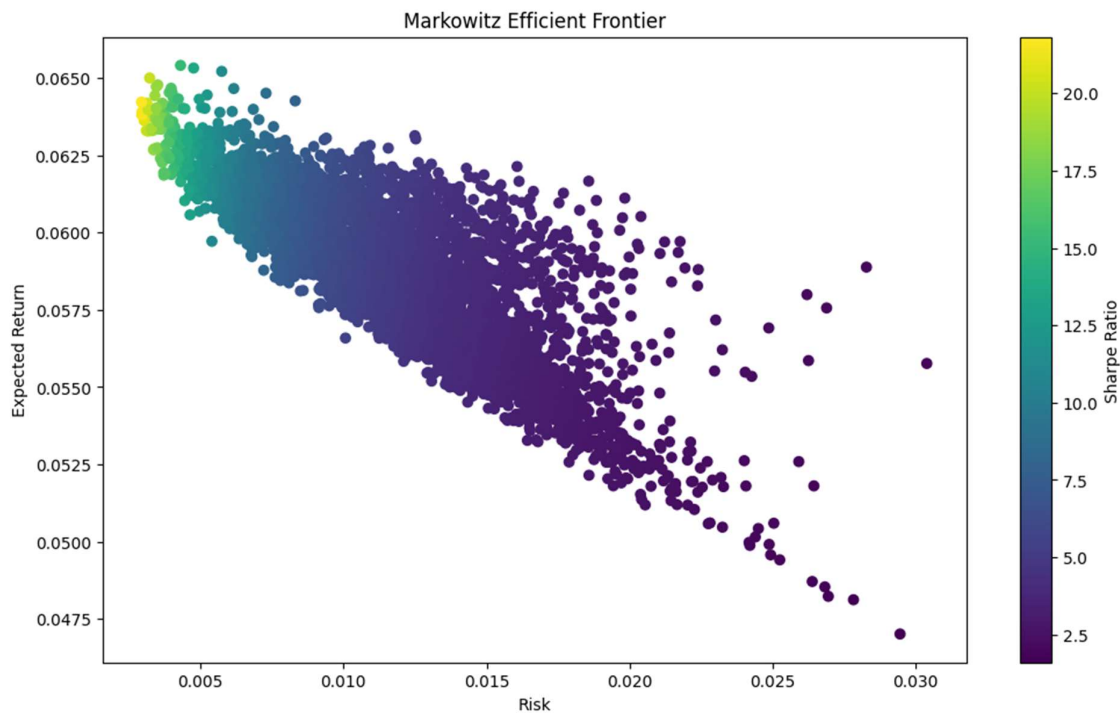
Objectives

- Maximize expected return.

- Minimize portfolio risk.
- Identify optimal asset allocation.

Results

Thousands of portfolio combinations were evaluated to identify efficient portfolios and maximize risk-adjusted returns.



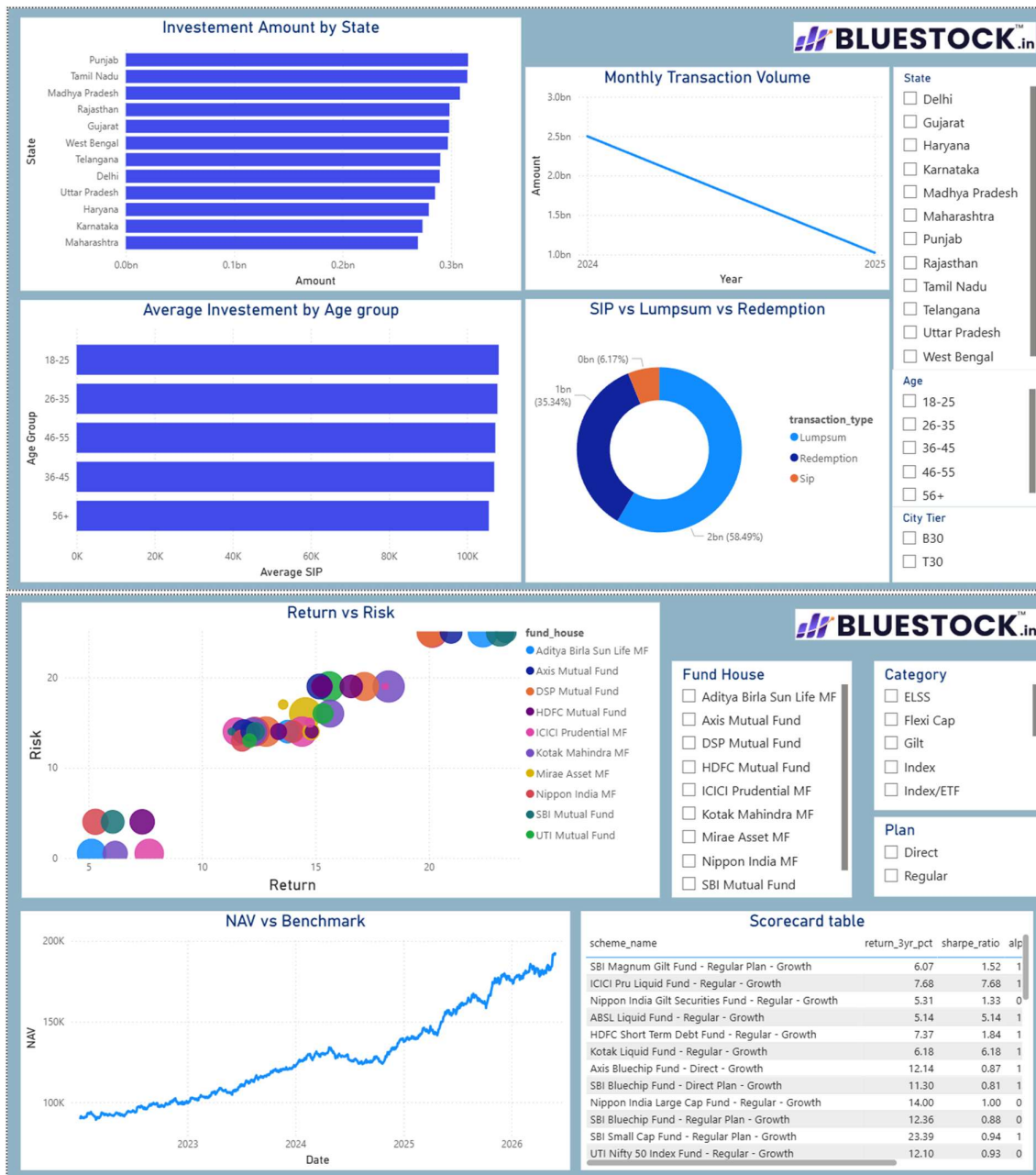
13. Dashboard Development

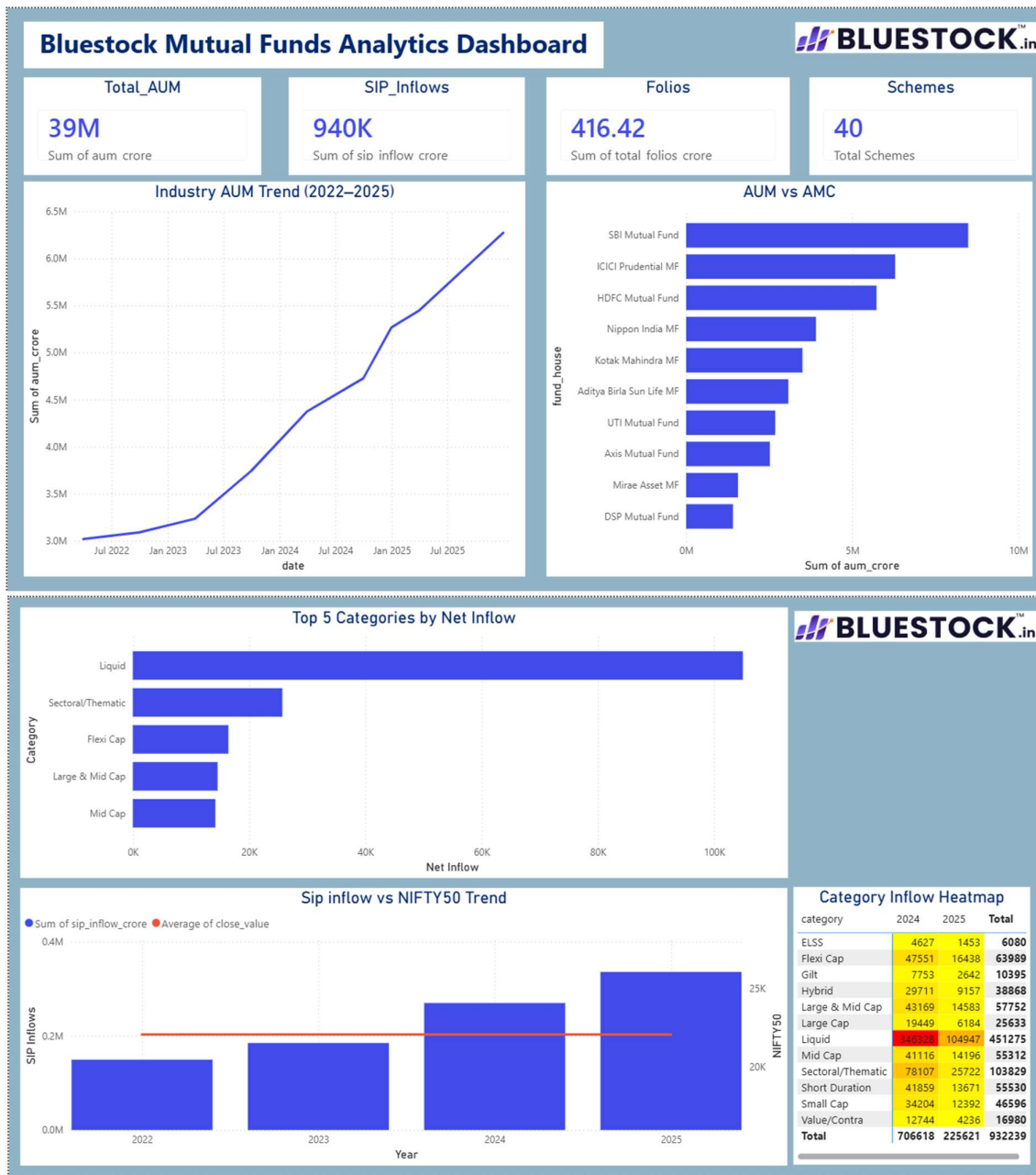
Power BI Dashboard

A four-page interactive dashboard was developed covering:

- Industry Overview
- Fund Performance
- Investor Analytics
- SIP & Market Trends

Features included filters, drill-through functionality, and interactive visualizations.





Streamlit Dashboard

A web-based dashboard was created using Streamlit to provide an alternative platform for fund analytics and recommendations.

Features include:

- KPI Cards
- Fund Performance Analysis
- Recommendation Engine

- Interactive Filtering



Return vs Risk Analysis

return_3yr_pct

25

20

15

10

5

0

std_dev_ann_pct

0

5

10

15

20

25

risk_grade

Moderate

Very High

Low

High

Moderately High

Fund Performance Table

	scheme_name	return_3yr_pct	sharpe_ratio	risk_grade
0	SBI Bluechip Fund - Regular Plan - Growth	12.36	0.88	Moderate
1	SBI Bluechip Fund - Direct Plan - Growth	11.3	0.81	Moderate
2	SBI Small Cap Fund - Regular Plan - Growth	23.39	0.94	Very High
3	SBI Small Cap Fund - Direct Plan - Growth	23.14	0.93	Very High
4	SBI Magnum Gilt Fund - Regular Plan - Growth	6.07	1.52	Low
5	HDFC Top 100 Fund - Regular Plan - Growth	14.84	1.06	Moderate
6	HDFC Top 100 Fund - Direct Plan - Growth	13.38	0.96	Moderate
7	HDFC Mid-Cap Opportunities Fund - Regular - Growth	16.58	0.87	High
8	HDFC Mid-Cap Opportunities Fund - Direct - Growth	15.29	0.8	High
9	HDFC Short Term Debt Fund - Regular - Growth	7.37	1.84	Low

Select Risk Grade

Moderate

	scheme_name	sharpe_ratio	return_3yr_pct
0	SBI Bluechip Fund - Regular Plan - Growth	0.88	12.36
1	SBI Bluechip Fund - Direct Plan - Growth	0.81	11.3
5	HDFC Top 100 Fund - Regular Plan - Growth	1.06	14.84
6	HDFC Top 100 Fund - Direct Plan - Growth	0.96	13.38
10	ICICI Pru Bluechip Fund - Regular - Growth	0.82	11.54
11	ICICI Pru Bluechip Fund - Direct - Growth	1.03	14.41
15	Nippon India Large Cap Fund - Regular - Growth	1	14
16	Nippon India Large Cap Fund - Direct - Growth	0.88	12.33
18	Nippon India ETF Nifty 50 BeES	0.91	11.77
20	Kotak Bluechip Fund - Regular - Growth	0.87	12.25

Top 10 Funds by Sharpe Ratio

	amfi_code	scheme_name	fund_house	category	plan	return_1yr_pct	return_3yr_pct	return_5yr_pct
14	120507	ICICI Pru Liquid Fund - Regular - Growth	ICICI Prudential MF	Liquid	Regular	8.89	7.68	7.5
23	120844	Kotak Liquid Fund - Regular - Growth	Kotak Mahindra MF	Liquid	Regular	4.26	6.18	8.2
30	101208	ABSL Liquid Fund - Regular - Growth	Aditya Birla Sun Life MF	Liquid	Regular	6.18	5.14	7.5
9	100025	HDFC Short Term Debt Fund - Regular - Growth	HDFC Mutual Fund	Short Duration	Regular	6.83	7.37	6.4
4	119120	SBI Magnum Gilt Fund - Regular Plan - Growth	SBI Mutual Fund	Gilt	Regular	5.34	6.07	5.4
19	118636	Nippon India Gilt Securities Fund - Regular - Growth	Nippon India MF	Gilt	Regular	6.19	5.31	8.7
34	148567	Mirae Asset Large Cap Fund - Regular - Growth	Mirae Asset MF	Large Cap	Regular	15.12	14.81	12.6
5	100016	HDFC Top 100 Fund - Regular Plan - Growth	HDFC Mutual Fund	Large Cap	Regular	10.94	14.84	11.3
11	120504	ICICI Pru Bluechip Fund - Direct - Growth	ICICI Prudential MF	Large Cap	Direct	14.12	14.41	13.0
15	118632	Nippon India Large Cap Fund - Regular - Growth	Nippon India MF	Large Cap	Regular	15.84	14	14

Fund Recommendation Engine

Select Risk Appetite

Moderate

Top 3 recommendations for Moderate risk investors

	scheme_name	sharpe_ratio	return_3yr_pct
5	HDFC Top 100 Fund - Regular Plan - Growth	1.0600	14.8400
34	Mirae Asset Large Cap Fund - Regular - Growth	1.0600	14.8100
11	ICICI Pru Bluechip Fund - Direct - Growth	1.0300	14.4100

Investor Insights

- Highest performing funds delivered strong risk-adjusted returns.
- Diversified funds exhibited lower drawdowns.
- SIP investors showed consistent long-term participation.
- High Sharpe funds dominated recommendation rankings.

14. Key Findings

- Industry AUM and SIP inflows demonstrated sustained growth.
- Equity-oriented funds generated higher returns with greater volatility.
- Risk-adjusted performance varied significantly across funds.
- Diversified portfolios exhibited lower concentration risk.
- Monte Carlo simulations highlighted uncertainty in future NAV growth.
- Portfolio optimization improved risk-return efficiency.

15. Limitations

- Limited historical data availability.
- Dependence on historical performance for forecasting.
- Simplified recommendation logic.

- Simulated investor data may not fully reflect real-world behaviour.
 - No real-time market integration.
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16. Recommendations

- Integrate real-time mutual fund data feeds.
 - Deploy dashboards on cloud infrastructure.
 - Enhance recommendation engine using machine learning.
 - Implement portfolio rebalancing suggestions.
 - Expand analytics with additional benchmark comparisons.
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17. Conclusion

The Bluestock Mutual Fund Analytics Platform successfully transformed raw mutual fund datasets into a comprehensive analytical ecosystem. Through ETL automation, database management, exploratory analysis, performance evaluation, advanced risk analytics, forecasting models, and interactive dashboards, the project demonstrates how data-driven techniques can support investment decision-making. The platform provides a scalable foundation for future enhancements and real-world deployment.